

Clinical Investigation

Dry Eye Syndrome After Proton Therapy of Ocular Melanomas



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Summary

Some institutions contraindicate proton therapy for ocular melanomas of superotemporal location owing to lacrimal gland irradiation and the risk of dry eye syndrome (DES). Severe DES is more frequent in temporal/superotemporal tumors. Severe DES correlates with temporal/superotemporal tumor location, anterior position, age, and retinal and lid irradiation. It involves the principal lacrimal gland and accessory lacrimal glands. Tumor location should not contraindicate proton therapy because it affected neither vision deterioration nor enucleation rate.

Purpose: To investigate whether proton therapy (PT) performs safely in superotemporal melanomas, in terms of risk of dry-eye syndrome (DES).

Methods and Materials: Tumor location, DES grade, and dose to ocular structures were analyzed in patients undergoing PT (2005-2015) with 52 Gy (prescribed dose, not accounting for biologic effectiveness correction of 1.1). Prognostic factors of DES and severe DES (sDES, grades 2-3) were determined with Cox proportional hazard models. Visual acuity deterioration and enucleation rates were compared by sDES and tumor locations.

Results: Median follow-up was 44 months (interquartile range, 18-60 months). Of 853 patients (mean age, 64 years), 30.5% had temporal and 11.4% superotemporal tumors. Five-year incidence of DES and sDES was 23.0% (95% confidence interval [CI] 19.0%-27.7%) and 10.9% (95% CI 8.2%-14.4%), respectively. Multivariable analysis showed a higher risk for sDES in superotemporal (hazard ratio [HR] 5.82, 95% CI 2.72-12.45) and temporal tumors (HR 2.63, 95% CI 1.28-5.42), age ≥ 70 years (HR 1.90, 95% CI 1.09-3.32), distance to optic disk ≥ 5 mm (HR 2.71, 95% CI 1.52-4.84), $\geq 35\%$ of retina receiving 12 Gy (HR 2.98, 95% CI 1.54-5.77), and eyelid rim irradiation (HR 2.68, 95% CI 1.49-4.80). The same risk factors were found for DES. Visual acuity deteriorated more in patients with sDES (0.86 ± 1.10 vs 0.64 ± 0.98 logMAR, $P = .034$) but not between superotemporal/temporal and other locations ($P = .890$). Enucleation rates were independent of sDES ($P = .707$) and tumor locations ($P = .729$).

Conclusions: Severe DES was more frequent in superotemporal/temporal melanomas. Incidence of vision deterioration and enucleation was no higher in patients with superotemporal melanoma than in patients with tumors in other locations. Tumor location should not contraindicate PT. © 2017 Elsevier Inc. All rights reserved.

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Conflict of interest: none.